

Binning criterion, proton-tagged angular observables and two-dimensional cross sections

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Studies of 2026-09-16/17; blind (fake data throughout)

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Abstract

Three questions were asked of the released analysis: whether the migration-diagonal requirement could be relaxed from 0.68 to 0.50, whether the proton-tagged sample should measure the proton angle and the pion-proton opening angle, and whether double-differential cross sections are supported. This note records the answers and the machinery built for them. **(i)** Relaxing the criterion admits finer binnings in $\cos\theta_\mu$, θ_μ , p_π and the proton-tagged transverse variables, but every one of the twenty-six candidates built and unfolded in full is rejected by the conditioning of the additional-smearing matrix A_C : the criterion is not what limits this measurement, and the study is reported as a study, with no rebinning adopted. **(ii)** θ_p (four bins) and the pion-proton opening angle $\theta_{\pi p}$ (three bins) both meet the unchanged 0.68 criterion and are now extracted for FHC, RHC and the combined configuration, with generator predictions from all four generators; $\theta_{\pi p}$ is the best-conditioned new observable of the analysis ($s_{\min}/s_1 = 0.30\text{--}0.48$). The selection now stores the opening angle, which required reprocessing the proton-tagged tree; the reprocess was gated on a bit-for-bit comparison of every existing branch. **(iii)** The framework supports two-dimensional measurements as flattened bins with one slice per outer bin; eight pairs were screened and are being extracted. The first results confirm what the one-dimensional study already showed: a good migration diagonal does not guarantee a usable A_C . The joint FHC+RHC background fit developed in the same period is documented separately in the control-region note.

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1 Scope, status and conventions

The signal region is blind: every “measurement” here is the per-run Poisson fake data of the released throw, and the extractions are closure tests. Three families of configuration are used and must not be confused.

- **Live:** θ_p and $\theta_{\pi p}$, in `configs/ccpilp*` with the released naming, extracted into `processed/` beside the released observables. They use binnings that meet the unchanged 0.68 criterion.
- **0.50 study:** `configs/c50/`, extracted into `processed/rebuild_c50/`. Nothing from it is adopted. The directory is deliberately outside the glob of `norm_manifest.py`, which audits the release.
- **2D study:** `configs/d2/`, extracted into `processed/rebuild_2d/`.

Screening uses the per-run exposure scales and the safe central-value weight of the extraction, and the same first/last-edge conventions as the released bin configs. Two criteria appear throughout: the column-normalised migration diagonal per true bin (the BNB criterion, > 0.68) and the conditioning of the additional-smearing matrix, $s_{\min}/s_1$ from its singular values, which the analysis note already identified as the quantity that decided $\cos \theta_\pi$ and $\theta_{\mu\pi}$.

2 The 0.50 migration-diagonal study

2.1 Screening

`macros/binning_screen_2d.C` evaluates every candidate in one pass over the processed files (the earlier `binning_screen.C` used four `TTree::Draw` calls per bin, which does not scale to a long candidate list). It reproduces the published pre-screen table exactly for FHC and RHC, which is its regression test, and it also handles two-dimensional candidates (§4). For the proton-tagged observables the maximin edge scan of `macros/tki_binning_scan.C` was generalised to take the criterion, an observable filter and the maximum bin count, and to cover W , the proton angles and the opening angle; it reproduces the two-bin transverse-kinematic-imbalance numbers of the binning README.

At 0.50 the following become admissible where 0.68 refused them (smallest diagonal, FHC / RHC): $\cos \theta_\mu$ with 6, 7 or 8 bins (0.64/0.63, 0.61/0.61, 0.60/0.59); θ_μ with 7 or 8 (0.63/0.62, 0.61/0.61); p_π with three bins and the extra edge at 0.26 GeV/c (0.51/0.53); proton-tagged δp_T with three (0.59/0.55), $\delta \alpha_T$ with three (0.52/0.50), $W_{\pi p}$ with two at 1.19 GeV/c (0.56/0.56); $\cos \theta_p$ with six and θ_p with seven. p_μ still fails (0.33/0.31 released, worse when finer), as do three-bin p_π at 0.30 GeV/c, four-bin p_n and $\delta \phi_T$ in RHC. W_{had} in two bins passes even at 0.68 (0.71/0.72) although the released configuration has six.

2.2 What the full extractions say

Twenty-six candidates (thirteen binnings $\times$ FHC and RHC) were built and unfolded with the released machinery. Table 1 gives the result. The conditioning is decisive: of the finer binnings only W_{had} (two bins) keeps an A_C with $s_{\min}/s_1$ above 0.5, and the two-bin $W_{\pi p}$ and the four-bin $\cos \theta_p$ in FHC are the only others above 0.1. Several candidates have $s_{\min}/s_1$ numerically zero to three decimals: their A_C maps every model onto a single shape, so the extraction cannot be read as a shape measurement, exactly the pathology the analysis note documented for θ_μ . Closure χ^2 values are uniformly small — which is the point: they cannot distinguish these cases, because a rank-deficient A_C makes data and prediction agree by construction.

Table 1: Candidates admitted by the 0.50 criterion, built and unfolded in full (fake data). A_C rows: range of the row sums; $s_{\min}/s_1$: conditioning of A_C ; data stat.: bin-averaged data-statistical term; unc.: range of the per-bin total uncertainty; closure: range of the per-bin ratio to the A_C -smeared truth. Rows in bold keep a usable A_C .

Candidate	Config	bins	A_C rows	$s_{\min}/s_1$	data stat.	unc. [%]	closure
$\cos \theta_\mu$ 6	FHC	6	0.40–0.52	0.110	17.0%	30–76	0.80–1.21
	RHC	6	0.09–0.44	0.085	15.8%	31–133	0.58–1.25
$\cos \theta_\mu$ 7	FHC	7	0.21–0.37	0.024	23.8%	31–143	0.51–1.30
	RHC	7	0.03–0.52	0.054	19.3%	33–121	0.71–1.17
θ_μ 7	FHC	7	0.47–0.68	0.000	11.0%	25–52	1.02–1.38
	RHC	7	0.35–0.93	0.001	10.3%	30–38	1.03–1.38
θ_μ 8	FHC	8	0.53–0.73	0.000	10.0%	25–54	0.87–1.36
	RHC	8	0.16–0.76	0.001	13.2%	28–54	0.84–1.42
p_π 3	FHC	3	0.53–0.91	0.048	14.8%	39–56	0.90–0.97
	RHC	3	0.45–0.89	0.010	12.7%	40–45	0.97–1.02
δp_T 3	FHC	3	0.37–0.89	0.001	13.8%	32–37	0.94–0.96
	RHC	3	0.32–0.86	0.000	12.0%	35–36	1.04–1.12
$\delta \alpha_T$ 3	FHC	3	0.44–0.98	0.029	10.2%	42–52	0.83–0.99
	RHC	3	0.42–0.92	0.011	9.8%	41–47	1.00–1.17
$W_{\pi p}$ 2 (1.19)	FHC	2	0.27–0.91	0.264	19.1%	41–72	0.93–0.99
	RHC	2	0.28–1.48	0.147	24.1%	47–91	0.73–1.00
W_{had} 2	FHC	2	0.80–0.90	0.579	16.3%	37–87	0.80–1.03
	RHC	2	0.76–0.86	0.541	17.1%	45–100	0.78–1.05
$\cos \theta_p$ 4	FHC	4	0.54–0.76	0.445	20.6%	34–76	0.66–1.08
	RHC	4	−0.09–0.62	0.018	36.2%	41–182	0.43–1.41
$\cos \theta_p$ 6	FHC	6	0.77–0.90	0.000	25.6%	34–84	0.54–1.06
	RHC	6	0.43–0.78	0.000	24.3%	38–72	0.75–1.48
θ_p 5	FHC	5	0.59–1.14	0.034	13.5%	33–44	0.84–1.24
	RHC	5	−0.03–0.60	0.005	24.9%	39–99	0.82–1.30
θ_p 7	FHC	7	0.61–0.98	0.000	16.4%	34–58	0.87–1.30
	RHC	7	0.32–0.87	0.000	20.7%	39–60	0.77–1.14

Conclusion. The 0.68 criterion is not the binding constraint of this analysis and lowering it to 0.50 buys nothing: the binnings it admits are rejected one level down, by the regularised response. No rebinning is proposed. Two by-products are worth keeping: W_{had} in two bins is better conditioned than the released six-bin configuration, and the criterion study produced the screening tool used for the new observables below.

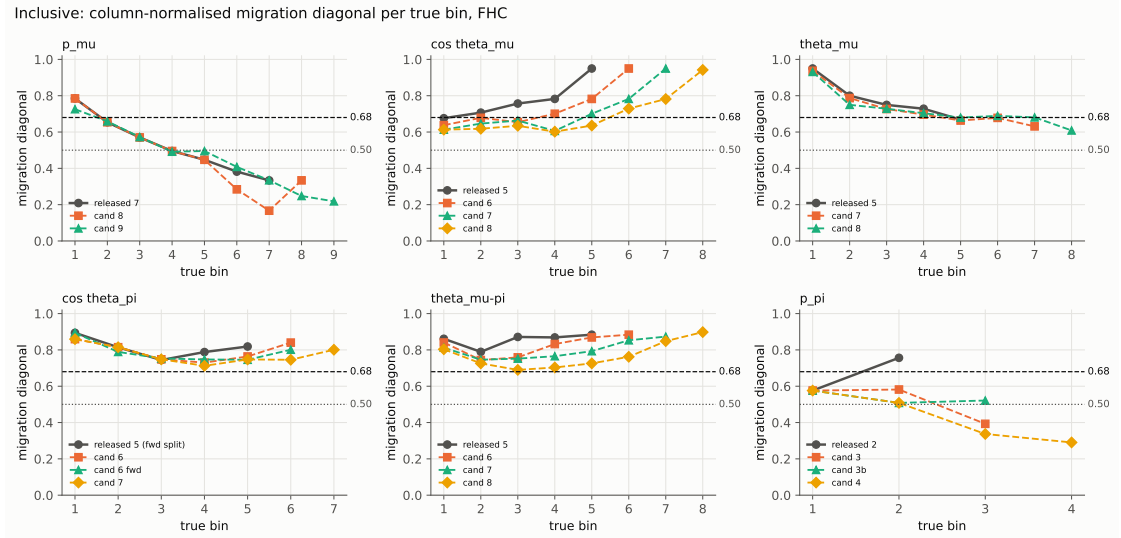


Figure 1: Migration diagonal per true bin for the released inclusive binnings and the finer candidates, FHC; the two criteria are drawn. The RHC version and the proton-tagged and two-dimensional versions are `c50/diag_{incl,1p,2d}_{fhc,rhc}`.

3 Proton-tagged angular observables

3.1 What was added, and why the opening angle needed a reprocess

The proton-tagged selection already stored the proton polar angle about the neutrino direction ($\cos \theta_p$, reconstructed in the beta convention, truth against the per-event neutrino). It did not store the pion–proton opening angle $\theta_{\pi p}$, nor the index of the chosen proton, so the angle could not be formed offline from the stored quantities. Both are now written by `CC1mu1pi1p`: the reconstructed angle from the two track directions (frame independent, and immune to the zero-momentum degeneracy of a scaled vector), the true angle from the leading charged pion and the leading proton.

Before committing to the reprocess, the proton choice was replicated offline (`macros/proton_angle_friend.C`) — the lowest-LLR track above threshold, contained, not the muon or pion candidate — and validated against the stored $\cos \theta_p$: identical for all 18 698 selected events of the fourteen Monte Carlo files. That friend tree also answered what the candidate is: on selected *signal* events it is the leading true proton 82% of the time, a softer proton 6%, and not a proton 13% (on all selected events, background included, 59, 10 and 31%). The wrong-candidate events populate a flat tail in both angles, while the correct ones peak within about 0.1 rad (figure `c50/resolution_proton_angles`).

Reprocess and its gate. The proton-tagged tree (41 units) and the seven per-run beam-off files, which are shared with the inclusive tree and carry both selections, were reprocessed into staging directories, and the per-run fake data re-thrown with the same seeds. Promotion was gated on `macros/compare_reprocess.C`: for every file, all 168 numeric branches of the old tree identical entry by entry, the entry count and `summed_pot` unchanged, and the new angle equal to the friend-tree value on every selected and signal event. All 56 files passed. One needed attention: the FHC Run-1 dirt snapshot carries a `summed_pot` written in place by `set_summed_pot.C`, 1.673920×10^{21} against the native 1.6739168×10^{21} ; the established value was carried into the new file so that the normalisation cache keys are unchanged. The old trees are kept as `w_pre_thpipr` and `ext_perrun_pre_thpipr`.

3.2 Binning

The maximin scan (FHC Runs 1, 2, 4; ≥ 400 events per bin) gives the largest bin count at each criterion; the weighted screen then checks it over FHC and RHC at the released exposure (Table 2). θ_p supports one bin more than $\cos \theta_p$ at both criteria, so the angle rather than its cosine is measured, matching the released θ_μ convention. Both adopted binnings meet 0.68; the RHC θ_p value sits exactly at it, as the released $\cos \theta_\mu$ does.

Table 2: Proton-tagged angular binnings. Scan: worst diagonal of the maximin scheme at that bin count (unweighted, FHC Runs 1, 2, 4). Screen: smallest diagonal FHC / RHC with the per-run weighting. Adopted binnings in bold.

Observable	bins	scan	screen FHC / RHC	edges
θ_p	4	72.6%	0.73/0.68	0, 0.597, 0.942, 1.287, π
θ_p	5	68.0%	0.67/0.64	0, 0.377, 0.691, 0.942, 1.225, π
θ_p	7	57.1%	0.56/0.55	(0.50 criterion)
$\cos \theta_p$	4	73.2%	0.73/0.66	−1, 0.275, 0.575, 0.825, 1
$\cos \theta_p$	6	57.2%	0.58/0.55	(0.50 criterion)
$\theta_{\pi p}$	3	75.8%	0.77/0.75	0, 1.193, 2.010, π
$\theta_{\pi p}$	4	67.9%	—	0, 0.973, 1.476, 2.010, π
$\theta_{\pi p}$	7	53.9%	0.52/0.53	(0.50 criterion)

3.3 Generator predictions

The four generator predictions are produced in `generator_predictions/gen2d/`, which holds copies of the released readers (GENIE `gst`, GiBUU `FinalEvents`, NuWro and NEUT in their containers) with the new observables added through one shared header, so the released chain is untouched. Two checks are built in. Every histogram the copies also write is reproduced bit-for-bit from the released per-flavour files, for all four generators; and the generic combiner reproduces all sixty released inclusive FTE histograms (four generators $\times$ FHC, RHC, combined $\times$ five observables) exactly. Unlike the released proton-tagged predictions, which exist for FHC only, the new ones are produced for FHC, RHC and the combined configuration. `scripts/check_ext_bins.py` checks that the generator edges and the configuration edges agree and that each prediction has as many bins as its configuration.

3.4 Extractions

Table 3: Extractions of the new observables (fake data). Columns as in Table 1; χ^2 is the closure against the A_C -smeared truth.

Observable	Config	bins	A_C rows	$s_{\min}/s_1$	data stat.	unc. [%]	closure χ^2/ndf
θ_p	FHC	4	0.58–1.30	0.082	12.3%	35–45	0.91/4
	RHC	4	0.21–1.00	0.002	12.9%	36–44	0.49/4
	COMB	4	0.60–1.24	0.125	9.7%	32–50	0.76/4
$\theta_{\pi p}$	FHC	3	0.57–0.65	0.480	20.1%	38–93	0.16/3
	RHC	3	0.59–0.73	0.376	24.5%	43–187	0.12/3
	COMB	3	0.43–0.49	0.304	14.3%	40–89	0.06/3

$\theta_{\pi p}$ is the best-conditioned observable of the proton-tagged programme: its $s_{\min}/s_1$ of 0.30–0.48 is above everything in Table 1 except two-bin W_{had} , and above the released $\cos\theta_\pi$ (0.27/0.24). Its price is the last bin, whose total uncertainty reaches 93% (FHC) and 187% (RHC). θ_p unfolds well in FHC and the combined configuration but is nearly rank-deficient in RHC (0.002), where its diagonal sat exactly at the criterion: the RHC θ_p curves should not be read as a shape measurement.

Proton-tagged CC1 μ 1 $\pi^\pm$: new observables, FAKE DATA (signal region blind); curves and integrated σ [10^{-38} cm 2 /Ar] are A_C -smeared

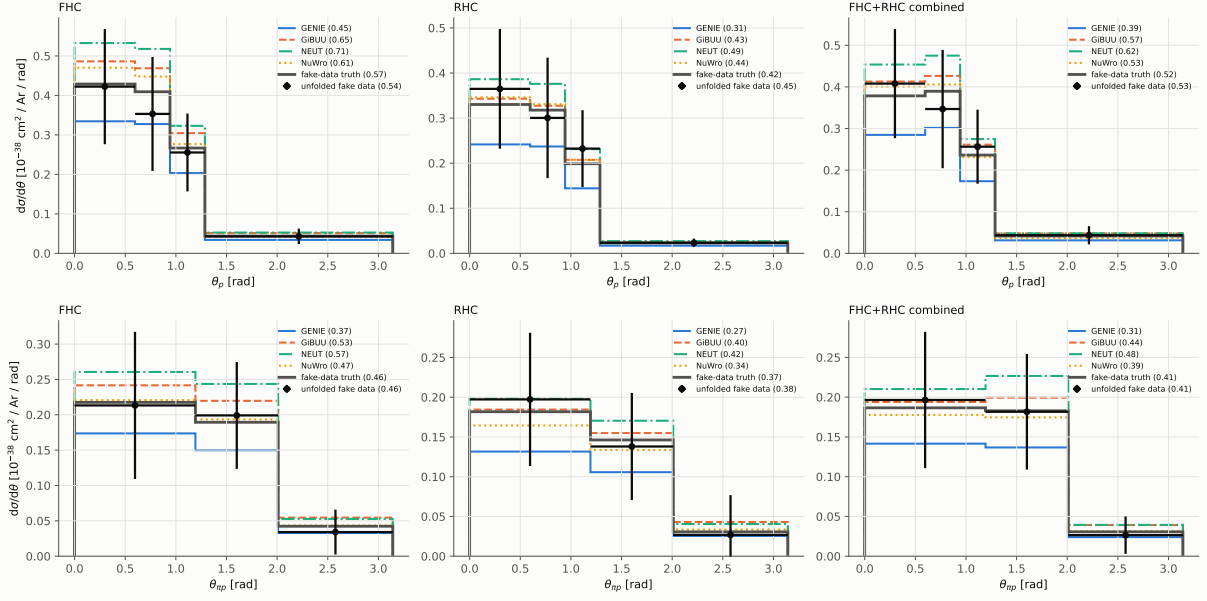


Figure 2: Extracted $d\sigma/d\theta_p$ and $d\sigma/d\theta_{\pi p}$ (fake data; the signal region is blind) against the four generators and the fake-data truth. Integrated cross sections in the legends, 10^{-38} cm 2 /Ar, A_C -smeared.

Both distributions are strongly forward peaked: the top θ_p bin, more than half the angular range, carries about a tenth of the first bin's $d\sigma/d\theta$, and the pion and proton are separated by less than 2 rad in the bulk of the events. The generator ordering is the familiar one — GENIE lowest, NEUT highest, GIBUU and NuWro between, a spread of about 1.6 in the integral — and the shapes are similar, so these observables discriminate mainly through normalisation, with NEUT's relative excess in the middle $\theta_{\pi p}$ bin the exception.

4 Two-dimensional cross sections

4.1 What the framework already supports

A double-differential measurement needs no new unfolding machinery. UniverseMaker bins are arbitrary selection strings, so a two-dimensional binning is a flat list of bins; **SliceBinning** carries slices with an “other variable” held in a fixed range, and **UnfolderNuMI** loops over the slices and divides by the width of the other variable, so each slice is a differential cross section in the inner variable. Two constraints were found and respected: the analysis bins of a slice must be contiguous, because the per-slice A_C block is taken as a range, so the bin configuration is written slice by slice; and the closure sidecar was dumped only for slice 0, which for a two-dimensional configuration is the first outer bin rather than the whole measurement, so **UnfolderNuMI** now also dumps the “bin number” slice as `closure_hists_all_*` when a configuration has more than one physical slice (one-dimensional output is unchanged). Configurations are generated by `scripts/make_bin_configs.py`, which reproduces the hand-written one-dimensional candidate configurations byte for byte as its regression test.

4.2 Screening

Eight pairs were screened with `binning_screen_2d.C`, which for a two-dimensional candidate also splits the off-diagonal migration into leakage within the outer slice (X) and between slices (Y). Table 4 gives the best grid of each pair. The Y-leakage is what decides: the pairs built on δp_T lose 20–30% of events across the outer bin and those built on $W_{\pi p}$ up to 60%, which is why

every $W_{\pi p}$ pair fails both criteria. Expected event counts are the second constraint: $\cos \theta_{\pi} \times \theta_{\mu\pi}$ has a bin with four expected events.

Table 4: Two-dimensional pairs (inner $\times$ outer), best grid of each: smallest diagonal over the flattened true bins, mean leakage within and between outer slices, and the expected selected signal of the thinnest bin at data exposure.

Pair	grid	min. diagonal FHC / RHC	X-leak	Y-leak	thinnest bin
$\cos \theta_{\pi} \times \cos \theta_{\mu}$	2×3	0.69/0.68	3.4%	12.5%	25
$\cos \theta_{\mu} \times p_{\mu}$	2×2	0.70/0.68	6.1%	11.5%	69
$\cos \theta_{\pi} \times \theta_{\mu\pi}$	2×3	0.66/0.60	5.7%	9.1%	4
$\cos \theta_{\pi} \times p_{\pi}$	2×2	0.53/0.51	5.4%	17.2%	34
$\theta_p \times \delta p_T$	2×2	0.53/0.52	8.4%	25.4%	63
$\theta_{\pi p} \times \delta p_T$	2×2	0.56/0.55	10.7%	25.4%	58
$\theta_p \times W_{\pi p}$	2×2	0.40/0.39	8.8%	39.4%	17
$\theta_{\pi p} \times W_{\pi p}$	2×2	0.21/0.20	9.7%	39.4%	12

4.3 Extractions

All eight pairs are being extracted for FHC, RHC and the combined configuration, with generator predictions produced as flattened histograms in the same bin order. At the time of writing the first three are complete and already repeat the lesson of §2: $\cos \theta_{\pi} \times \cos \theta_{\mu}$ gives $s_{\min}/s_1 = 0.245$ (FHC) and 0.315 (RHC) with a data-statistical term of 29 and 25%, while $\cos \theta_{\mu} \times p_{\mu}$ — the pair with the best migration diagonals of the set — comes out at 0.002 in FHC, rank-deficient. A good diagonal does not imply a usable regularised response. The completed table will be added when the array finishes; the evaluation is `report/tools/extract_eval.py`, which reads the multi-slice sidecar where one exists.

5 Reproduction

```
# screening (single pass, 1D and 2D; reproduces the published pre-screen exactly)
root -l -b -q 'macros/binning_screen_2d.C+("fhc","incl")'          # rhc; incl | 1p
root -l -b -q 'macros/tki_binning_scan.C("/data/uboone/processed/w",\
                                     400,0.68,"thetap,thpipr",10)'

# theta_pipr without reprocessing (friend trees, for screening only)
logs/c50/run_pafriend.sh
# configurations
python3 scripts/make_bin_configs.py c50_candidates                # configs/c50
python3 scripts/make_bin_configs.py newobs_candidates             # configs/ + configs/d2
python3 scripts/check_ext_bins.py                                # generator vs configuration edges
# generator predictions (verifies the released histograms bit-for-bit)
generator_predictions/gen2d/regen_ext.sh
# reprocess with the opening angle, its gate, and promotion
./reprocess_thpipr.sh ; PROMOTE=1 ./reprocess_thpipr.sh
# extractions
sbatch --array=0-25%4 --export=ALL,REPO=$PWD,\
      MANIFEST=$PWD/slurm/c50_manifest.list    slurm/slurm_c50.sbatch
sbatch --array=0-29%6 --export=ALL,REPO=$PWD,\
      MANIFEST=$PWD/slurm/newobs_manifest.list slurm/slurm_extract.sbatch
# evaluation and figures
python3 report/tools/binning_eval_c50.py    # 0.50 candidates
python3 report/tools/extract_eval.py        # new observables and 2D
python3 report/tools/c50_figures.py
python3 report/tools/newobs_xsec_fig.py
```

6 Open items

1. The two-dimensional extractions are running; their conditioning decides whether any pair is worth proposing as a released measurement. On the evidence so far only $\cos \theta_\pi \times \cos \theta_\mu$ is a candidate, and its data-statistical term is large.
2. θ_p in RHC is rank-deficient. Either it is quoted for FHC and the combined configuration only, or the RHC binning is coarsened to three bins, which has not been screened.
3. Two-bin W_{had} is better conditioned than the released six-bin configuration; a decision on replacing it is outstanding.
4. The proton candidate is the leading true proton only 82% of the time on signal events, and the truth-level observables use the leading proton. The effect is inside the response, but an inverted-proton-PID control region — already a prerequisite for the signal region — is what would constrain it.
5. The released analysis note does not yet mention θ_p or $\theta_{\pi p}$; the binning appendix and the observable tables need entries once the decisions above are taken.